MNIST is 'Hello World' to machine learning

Mixed National Institute of Standards and Technology (MNIST) database is the computer vision dataset that is used to train the machine learning system. It is basically a set of handwritten digits that the system has to learn and identify by the corresponding label. Accuracy of your model will depend on the intensity of your training. Broader the training data set, better will be the accuracy of your model.

One example is Softmax Regression model, which exploits the concept of probability to decipher a given image. As every image in MNIST is a handwritten digit between zero and nine, the image you are analysing can be only one of the ten digits. Based on this understanding, the principle of Softmax Regression allots a certain probability of being a particular number, to every image under test.

Smart handling of resources. As this process might involve a good bit of heavy lifting, just like other compute-heavy operations, TensorFlow offloads the heavy lifting outside Python environment. As the developers describe it, instead of running a single expensive operation independently from Python, TensorFlow lets you describe a graph of interacting operations that run entirely outside Python.

A few noteworthy features

Using TensorFlow to train your system comes with a few added benefits.

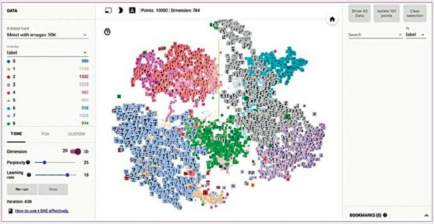

Fig. 2: Embeddings in TensorFlow

Visualising learning. No matter what you hear or read, it is only when you visually see something that the concept stays in your mind. The easiest way to understand the computational graph is, of course, to understand it pictorially. A utility called TensorBoard can display this very picture. The representation is very similar to a flow or a block diagram.

Graph visualisation. Computational graphs are complicated and not easy to view or comprehend. The graph visualisation feature of TensorBoard helps you understand and debug the graphs easily. You can zoom in or out, click on blocks to check their internals, check how data is flowing from one block to another and so on. Name your scopes as clearly as possible in order to visualise better.

The graph also includes series collapsing—names with number indexes are condensed to a single line on the graph, which can be expanded at your will. There are also special icons for different types of nodes to help you distinguish these easily. You can dump statistical information like time and resource usage data if you need.

Other than viewing the graphs pictorially, TensorFlow lets you plot quantitative metrics about the execution of your graphs and also shows additional data like images that pass through it. The base for TensorBoard is the event file generated while running the tool. This file contains summary data from nodes that you select for generating a summary.

Embedding visualisation. Embedding Projector, the built-in visualiser of TensorFlow, aids interactive visualisation and analysis of high-dimensional data like embeddings. The projector reads the embeddings from a model checkpoint file and loads any 2D tensor or embeddings.

Begin with the smallest unit, the tensor

Data in TensorFlow is based on its central unit, a tensor. A tensor is an array of a primitive set of values, and can have any number of dimensions.

Loading data. Sending data into a TensorFlow program can be

A thing or two that might help

- Background in Python programming, machine learning and working with arrays will help you use the tool better.
- The project can have nodes as data queues.
- Multiple threads can be handled easily by Coordinator class.
- Place pre-processing operations on the CPU instead of the GPU to speed up the process.
- TensorFlow debugger lets you see the internal structure and states if running TensorFlow graphs. This comes in handy while debugging various types of model bugs during training and inference.
- Some other stuff you can use TensorFlow for, apart from machine learning:
  - Simulate the behaviour of a partial differential equation
  - Visualise Mandelbrot Set